

Nishanth Kunchala

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Education

The Ohio State University

Aug 2022 – May 2026

BS in Aerospace Engineering, Minor in German

- GPA: 3.9/4.0
- **Coursework:** Flight Vehicle Structures, Flight Vehicle Controls, Flight Vehicle Dynamics, Space Vehicle Design, Hypersonic Flow, Computational Fluid Dynamics, Heat Transfer, Principles of Propulsion

Goethe-Institut Dresden

June 2024

B2 Course Certificate

- Completed the Goethe I75 German course at the CEFR B2 level, during a summer study abroad program with a heavy emphasis on cultural immersion and language practice experiences

Experience

NASA Glenn Research Center Internship

Cleveland, OH

Aircraft Design and Optimization in Aviary

June 2025 – Aug 2025

- Improved Aviary's utility for Design-Build-Fly teams by introducing **small-scale UAV conceptual design** capabilities—previously unavailable—through custom mass and geometry subsystems, as well as **CFD mesh generation and visualization via OpenVSP**
- Developed an analytical structural mass and geometry subsystem with OpenVSP integration, **achieving mass estimates within 2.5% of actual values** while minimizing required inputs to enable more accurate results and streamlined optimization workflows
- Built trajectory optimization workflows in Dymos and OpenMDAO to generate optimal aircraft designs, **increasing mission range by 21%**, through collaboration with propulsion and aerodynamic teams, to minimize mass and power consumed, while optimizing lift

The Ohio State University

Columbus, OH

Teaching Assistant & Grader

Jan 2024 – Present

- Graded assignments and exams for over 130 students in **AEROENG 3522: Astronautics**, developing detailed rubrics and providing constructive, individualized feedback on orbital mechanics assignments and projects
- Instructed and mentored 24 students in **AEROENG 2200: Introduction to Aerospace Engineering**, leading biweekly lab and recitation sessions focused on aerospace fundamentals and MATLAB-based analysis
- Graded assignments and exams for over 140 students in **MECHENG 2040: Statics and Introduction to Mechanics of Materials**, developing detailed rubrics and providing constructive, individualized feedback on static equilibrium and stress analysis

Projects

Senior Experimental Capstone

Aug 2025 – May 2026

- Designed a novel **Cycling PWPF-modulated LQR control allocation scheme** for an over-actuated, shared-tank cryogenic 1U CubeSat using **cold-gas CO₂ propulsion**, enabling precise maneuvering while preventing simultaneous actuation on shared fuel lines
- Evaluated control methodology and actuator dynamics using high-fidelity physics-based simulations in **NVIDIA Isaac Sim**, validating controller stability and tracking accuracy against a baseline model
- Achieved a **70% reduction in propellant** consumption and bounded steady-state **pointing error to within $\pm 0.015^\circ$** , with findings accepted for publication and presentation at the 2026 AIAA Region III Student Conference

Senior Design Capstone

Aug 2025 – May 2026

- Designed an autonomous Mars logistics orbiter intended to establish a 10-20 year orbital presence to provide rapid-response resupply capabilities to future Martian surface installations via SpaceX's Starship.
- Architected a **telemetry and communications network** relying on an S-Band Medium Gain Parabolic Grid Antenna for Deep Space Network (DSN) integration, alongside an Electra UHF transceiver for surface-to-orbit proximity data relay.
- Integrated a **Guidance, Navigation, and Control (GNC)** suite for high-precision pointing and designed the descent payloads with a Flush Air Data System (FADS) and a radar altimeter to ensure optimal atmospheric entry performance.
- Performed **stress, buckling, and modal FEA** in **ANSYS Static Structural** on the orbiter chassis and solar arrays, achieving natural frequencies of **15.4 Hz lateral / 38.2 Hz axial** on the primary structure and **43.2 Hz** on stowed solar panels, exceeding SpaceX Falcon User's Guide requirements with margin, while maintaining a minimum structural safety factor of **2.07** under max launch loads.

NASA Human Lander Challenge (HuLC)

Jan 2025 – June 2025

- Validated microgravity cryogenic fuel transfer models with ANSYS Fluent CFD, **reducing mass flow rate loss to 0.24%**, coauthoring a technical paper on CFD-based autonomous coupling system design
- Programmed a 3D keypoint detection multi-headed neural network for in-orbit alignment using TensorFlow, achieving **12 ms inference speed on a 10,000-image dataset** by designing procedures for data collection, preprocessing, training, and optimization
- Engineered and programmed a 6-DOF Stewart platform for **autonomous cryogenic propellant coupling**, achieving **sub-millimeter positioning accuracy** by integrating 3D printing, linear actuators, **AprilTag and LiDAR-based visual feedback**, spider boards, and a custom **Arduino-based control loop**
- Presented work on autonomous coupling and cryogenic fuel transfer at the **2025 NASA HuLC Conference**, highlighting applications to future autonomous space systems

Research

OSU Aerospace Research Center

Columbus, OH

Drone-Based Wildfire Detection

Jan 2025 – Present

- Developed and optimized a novel Context-Aware Centralized Copy-Paste Data Augmentation (CCPDA) pipeline, **increasing usable wildfire dataset size by 20×** and enabling robust wildfire segmentation under data scarcity
- Achieved a U-Net segmentation **accuracy increase from 40% to 97%** on a **400M-sample, 4-class wildfire dataset** by applying custom loss functions based on firefighter response priorities and performing hyperparameter tuning with cross-validation across 48 model configurations
- Engineered a fire-cluster pasting algorithm using contextual semantic statistics and similarity metrics, **reducing False Negative Rate by 26.6%** on real-world wildfire data

Publications

Performance of a Cycling PWPF Control Allocation Scheme for Limited Mass Flow Propulsion Systems

Apr 2026

Nishanth Kunchala, Antonio Accardo, Max Cirino, Brody Booker, Varun Chitiveli

- Accepted for publication in the proceedings of the 2026 AIAA Region III Student Conference, Ann Arbor, MI.

Design, Testing, and Comparisons of a Carbon Dioxide Compressed Gas Propulsion Module for 1U CubeSat Platforms

Apr 2026

Varun Chitiveli, Brody Booker, Max Cirino, Antonio Accardo, **Nishanth Kunchala**

- Accepted for publication in the proceedings of the 2026 AIAA Region III Student Conference, Ann Arbor, MI.

Centralized Copy-Paste: Enhanced Data Augmentation Strategy for Wild-land Fire Semantic Segmentation

Jan 2026

Joon Tai Kim, Tianle Chen, Ziyu Dong, **Nishanth Kunchala**, Alexander Guller,
Daniel Ospina Acero, Roger Williams, Mrinal Kumar

[10.48550/arXiv.2507.06321](https://arxiv.org/abs/10.48550/arXiv.2507.06321) [↗](#)

- Accepted for publication in the proceedings of the AIAA SciTech 2026 Forum, Orlando, FL.

Certifications

ANSYS Systems Tool Kit Level 1 Certification	<i>Sep 2025</i>
CEFR – Level B2 German Language Certification	<i>June 2024</i>
PCEP – Certified Entry-Level Python Programmer	<i>April 2024</i>

Awards

2nd Place, AIAA Region III Student Conference	<i>April 2026</i>
Best Prototype and \$9,250 in prototype funding, NASA’s HuLC Finalist Conference	<i>June 2025</i>
3rd Place, OSU HackAI Hackathon	<i>Feb 2025</i>
2nd Place, Ohio State Engineering FYE Design Showcase	<i>April 2023</i>

Technical Skills

Programming Languages: Python, MATLAB, Java, C, LaTeX, Javascript, HTML

Simulation & AI Tools: OpenMDAO, Dymos, ANSYS Fluent, ANSYS STK, IsaacSim, TensorFlow, Simulink

CAD & Design: SOLIDWORKS, OpenVSP, Onshape

Workflow & Documentation: Git, Overleaf, Microsoft Office